

1 Solution — GIAM 3.1

Problem 7

1.1 Problem

Prove that if x is an odd integer, then x^2 is of the form $4k + 1$ for some integer k .

1.2 Proof

***Proof.** Let x be an odd integer. By the definition of odd there is an integer m with $x = 2m + 1$. Squaring,*

$$x^2 = (2m + 1)^2 = 4m^2 + 4m + 1 = 4(m^2 + m) + 1.$$

Now $m^2 + m$ is an integer, since $\mathbb{Z}$ is closed under multiplication and addition. So $x^2 = 4k + 1$ with $k = m^2 + m \in \mathbb{Z}$. ■

The one idea is to expand and then **group so that a factor of 4 comes out front**, leaving exactly the $+1$ the conclusion demands.

1.3 The Proof Gives More Than Was Asked

Expand the square, then group so that a factor of 4 can be pulled out front.

UNPACK, COMPUTE, REPACK

x is odd $\Rightarrow x = 2m + 1, m \in \mathbb{Z}$

$$x^2 = (2m + 1)^2 = 4m^2 + 4m + 1 = 4(m^2 + m) + 1$$

$m^2 + m \in \mathbb{Z}$, so $x^2 = 4k + 1$ with $k = m^2 + m$

that is the theorem ■

But look again at k — the proof gives more than was asked

$$k = m^2 + m = m(m + 1)$$

a product of two *consecutive* integers — one of them must be even, so k is even

$$\text{write } k = 2j. \text{ Then } x^2 = 4(2j) + 1 = 8j + 1$$

every odd square is 1 more than a multiple of 8, not merely of 4

The first few odd squares

x	x^2	as $4k + 1$	as $8j + 1$
1	1	$4(0) + 1$	$8(0) + 1$
3	9	$4(2) + 1$	$8(1) + 1$
5	25	$4(6) + 1$	$8(3) + 1$
7	49	$4(12) + 1$	$8(6) + 1$
9	81	$4(20) + 1$	$8(10) + 1$

note the k column is always even — that is $m(m+1)$ at work

Verified for every odd x in $[-99, 99]$: $x^2 \equiv 1 \pmod{4}$ and also $x^2 \equiv 1 \pmod{8}$.

Figure 1.1: Top panel, UNPACK COMPUTE REPACK: x odd gives x equal to $2m$ plus 1 with m an integer; then x squared equals $(2m$ plus 1) squared equals $4m$ squared plus $4m$ plus 1 equals 4 times $(m$ squared plus $m)$ plus 1; and m squared plus m is an integer, so x squared equals $4k$ plus 1 with k equal to m squared plus m — that is the theorem. Second panel: but look again at k , since the proof gives more than was asked. k equals m squared plus m equals m times $(m$ plus 1), a product of two consecutive integers, one of which must be even, so k is even. Writing k as $2j$ gives x squared equals 4 times $2j$ plus 1, that is $8j$ plus 1 — so every odd square is one more than a multiple of 8, not merely of 4. Below, a table of the first few odd squares: 1 squared is 1, which is 4 times 0 plus 1 and 8 times 0 plus 1; 3 squared is 9, which is 4 times 2 plus 1 and 8 times 1 plus 1; 5 squared is 25, which is 4 times 6 plus 1 and 8 times 3 plus 1; 7 squared is 49, which is 4 times 12 plus 1 and 8 times 6 plus 1; 9 squared is 81, which is 4 times 20 plus 1 and 8 times 10 plus 1. The k column is always even, which is m times m plus 1 at work. Verified for every odd x in the range minus 99 to 99, both mod 4 and mod 8.

Look closely at the witness the proof produced:

$$k = m^2 + m = m(m + 1),$$

a product of two **consecutive** integers. One of any two consecutive integers is even, so k is always even. Writing $k = 2j$,

$$x^2 = 4(2j) + 1 = 8j + 1.$$

So the proof establishes the strictly stronger fact that **every odd square is 1 more than a multiple of 8**. Verified for every odd x in $[-99, 99]$.

1.4 Remark — Read Your Own Witness

It is worth pausing on how that strengthening was found: not by a new idea, but by **looking at the object the proof already produced**. The conclusion only asked for " k is an integer," so it is tempting to check that box and stop. But k came out in the recognisable form $m(m+1)$, and recognising it costs one line.

This is a generally useful habit. A constructive proof hands you a specific witness, and that witness often satisfies more than the theorem demanded. Asking "what else is true of the thing I just built?" is one of the cheapest ways to strengthen a result.

1.5 Remark — Squares Mod 4: Only Two Residues Occur

The companion computation for even x completes the picture. If $x = 2m$ then $x^2 = 4m^2 \equiv 0 \pmod{4}$. So

$$x \text{ even} \implies x^2 \equiv 0, \quad x \text{ odd} \implies x^2 \equiv 1 \pmod{4},$$

and **no perfect square is ever 2 or 3 modulo 4** (confirmed over $[-99, 99]$). That negative fact is the useful one in practice: it lets you rule out solutions instantly. For instance $x^2 + y^2 = 3$ has no integer solutions, since the left side can only be 0, 1 or 2 mod 4 while $3 \equiv 3$; and no integer of the form $4n + 3$ is a perfect square.

1.6 Remark — This Refines the Parity Lemma

Problem 2.1.3 established the lemma " x odd $\implies x^2$ odd," proved by expanding $(2k+1)^2 = 2(2k^2 + 2k) + 1$. The present theorem is the *same computation grouped one step further*: instead of pulling out a 2, pull out a 4.

$$(2m + 1)^2 = \underbrace{2(2m^2 + 2m) + 1}_{\text{gives: odd}} = \underbrace{4(m^2 + m) + 1}_{\text{gives: } \equiv 1 \pmod{4}} = \underbrace{8(j) + 1}_{\text{gives: } \equiv 1 \pmod{8}}$$

Each regrouping extracts more information from the identical algebra. The first version was exactly what the $\sqrt{2}$ proof of Section 1.6 needed; the sharper versions matter elsewhere, and the mod-8 form is what drives the classical proof that no integer of the form $8n + 7$ is a sum of three squares.

1.7 Remark — The Converse Holds Too

Combining with the parity lemma gives a chain of equivalences:

$$x \text{ is odd} \iff x^2 = 4k + 1 \iff x^2 = 8j + 1.$$

The rightward implications are this problem. For the leftward direction, $x^2 = 4k + 1$ makes x^2 odd, and Problem 2.1.3's lemma then forces x odd. So although the problem is stated as a one-way conditional, all three conditions are interchangeable. As in Problem 5, it is worth checking whether a converse comes for free — here it does, but only because a previously proved lemma is available to supply it.

1.8 Remark — Why $m(m + 1)$ Is Always Even

The step “one of two consecutive integers is even” deserves its own justification, since the whole strengthening rests on it. By the quotient-remainder theorem with divisor 2, either $m = 2t$ or $m = 2t + 1$:

- if $m = 2t$, then $m(m + 1) = 2t(2t + 1)$, visibly even;
- if $m = 2t + 1$, then $m + 1 = 2t + 2 = 2(t + 1)$, so $m(m + 1) = 2(t + 1)(2t + 1)$, also even.

This is a genuine proof by cases (Problem 2.6.2), and it generalises: the product of any n consecutive integers is divisible by $n!$ — which is the same fact that makes every binomial coefficient an integer.